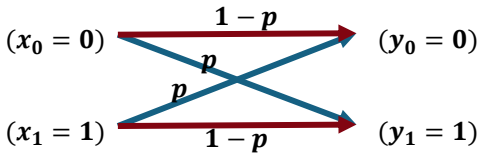
Cairo University
Faculty of Engineering
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Information Theory
3rd year 2026 – 2027

Sheet 1

Problem 1:

Consider the given binary symmetric channel:
Let $\mathbf{p_0}$ denote the probability of sending binary symbol $\mathbf{x_0}$. Let $\mathbf{p_1 = 1 - p_0}$ denote the probability of sending binary $\mathbf{x_1}$ Let $\mathbf{p}$ denote the transition probability of the channel.



a) Show that **the mutual information** between the channel input & channel output is given by:

$$I(X;Y) = H(Z) - H(p)$$

where:

$$H(z) = z\log_2\left(\frac{1}{z}\right) + (1-z)\log_2\left(\frac{1}{1-z}\right)$$

$$z = p_0 p + (1-p_0)(1-p),$$

and:

$$H(p) = p\log_2\left(\frac{1}{p}\right) + (1-p)\log_2\left(\frac{1}{1-p}\right)$$

b) Show that the value of p_o **that maximizes** $I(X;Y)$ **is equal to** $1/2$

c) Show that the **channel capacity** equals

$$C = 1 - H(p)$$

Problem 2:

A Binary Channel Matrix is Given by

$$P(Y|X) = \begin{array}{c} \\ x_0 \\ x_1 \end{array} \begin{array}{c} \begin{array}{cc} y_0 & y_1 \end{array} \\ \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{9}{10} \end{bmatrix} \end{array}$$

$$P(x=0) = \frac{1}{3}, P(x=1) = \frac{2}{3}$$

a) Determine $H(X),\ H(X \mid Y), H(Y),\ H(Y \mid X)$

b) Compute $I(X;Y)$ **(Mutual Information)**